Exp No:16 Experiment to understand the data preprocessing in Data Science

AIM:

To understand the data preprocessing in Data Science

Description:

Understand the importance of Data preprocessing in data science

CODE:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
df = pd.read_csv("/content/student_eda_dataset.csv")
df
df.head()
sns.displot(df.Marks, kde=True)
sns.displot(df.Marks, kde=False)
sns.jointplot(x=df.Study_Hours, y=df.Marks)
sns.jointplot(x=df.Study_Hours, y=df.Marks, kind="reg")
sns.jointplot(x=df.Study_Hours, y=df.Marks, kind="hex")
sns.pairplot(df[['Study_Hours', 'Attendance', 'Marks', 'Assignment_Score']])
sns.pairplot(
  df[['Study_Hours', 'Attendance', 'Marks', 'Assignment_Score', 'Study_Mode']],
  hue='Study_Mode'
)
sns.pairplot(
  df[['Study_Hours', 'Attendance', 'Marks', 'Assignment_Score', 'Gender']],
```

```
 hue='Gender'
)
sns.heatmap(
   df[['Study_Hours', 'Attendance', 'Marks', 'Assignment_Score']].corr(),
   annot=True
)
sns.boxplot(y=df.Marks)
sns.boxplot(y=df.Study_Hours)
sns.countplot(x=df.Department)
sns.countplot(x=df.Gender)
df.Gender.value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.ylabel('')
plt.show()
df.Gender.value_counts().plot(kind='bar')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.show()
sns.countplot(x=df.Passed)
sns.countplot(x=df.Study_Mode)
sns.countplot(x=df.Gender, hue=df.Passed)
sns.countplot(x=df[df.Study_Mode == 'Online']['Department'])
```

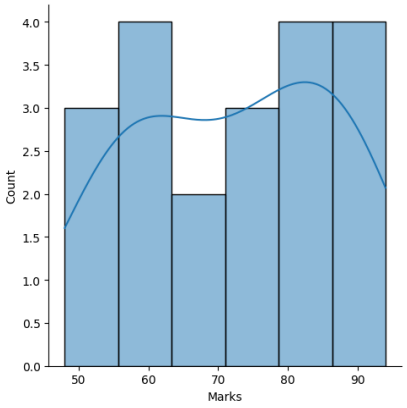
OUTPUT:

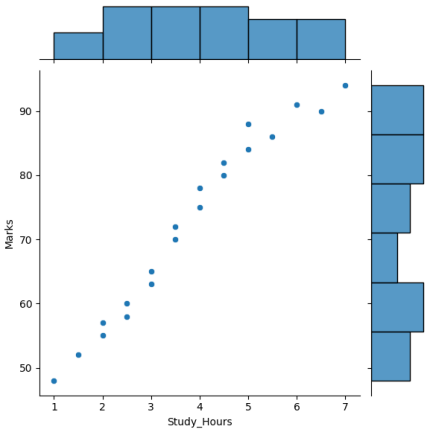
	Student_ID	Gender	Department	Study_Mode	Study_Hours	Attendance	Marks	Assignment_Score	Passed
0	101	Male	CSE	Online	2.5	72	58	60	No

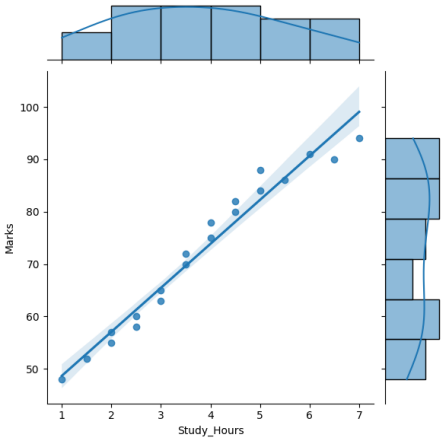
	Student_ID	Gender	Department	Study_Mode	Study_Hours	Attendance	Marks	Assignment_Score	Passed
1	102	Female	ECE	Offline	4.0	85	78	80	Yes
2	103	Male	CSE	Online	3.0	78	65	68	Yes
3	104	Female	IT	Offline	5.0	92	88	90	Yes
4	105	Male	ECE	Online	1.5	68	52	55	No
5	106	Female	CSE	Offline	6.0	95	91	93	Yes
6	107	Male	IT	Online	3.5	80	70	72	Yes
7	108	Female	ECE	Offline	4.5	88	82	85	Yes
8	109	Male	CSE	Online	2.0	70	55	58	No
9	110	Female	IT	Offline	5.5	90	86	88	Yes
10	111	Male	ECE	Offline	7.0	96	94	96	Yes
11	112	Female	CSE	Online	3.0	76	63	65	Yes
12	113	Male	IT	Offline	4.0	84	75	78	Yes
13	114	Female	ECE	Online	2.5	73	60	62	Yes
14	115	Male	CSE	Offline	6.5	94	90	92	Yes
15	116	Female	IT	Online	1.0	65	48	50	No

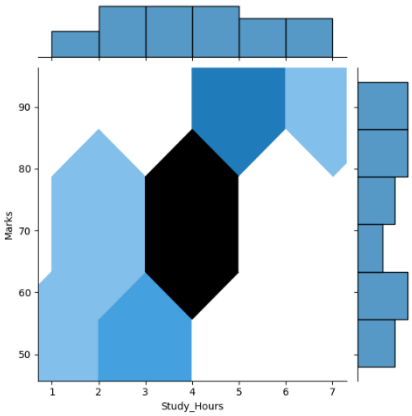
	Student_ID	Gender	Department	Study_Mode	Study_Hours	Attendance	Marks	Assignment_Score	Passed
16	117	Male	ECE	Offline	5.0	89	84	87	Yes
17	118	Female	CSE	Online	3.5	82	72	75	Yes
18	119	Male	IT	Offline	2.0	71	57	59	No
19	120	Female	ECE	Online	4.5	87	80	83	Yes

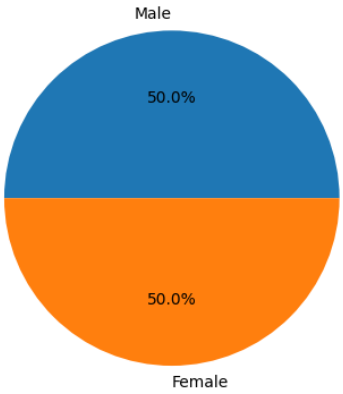
	Student_ID	Gender	Department	Study_Mode	Study_Hours	Attendance	Marks	Assignment_Score	Passed
0	101	Male	CSE	Online	2.5	72	58	60	No
1	102	Female	ECE	Offline	4.0	85	78	80	Yes
2	103	Male	CSE	Online	3.0	78	65	68	Yes
3	104	Female	IT	Offline	5.0	92	88	90	Yes
4	105	Male	ECE	Online	1.5	68	52	55	No

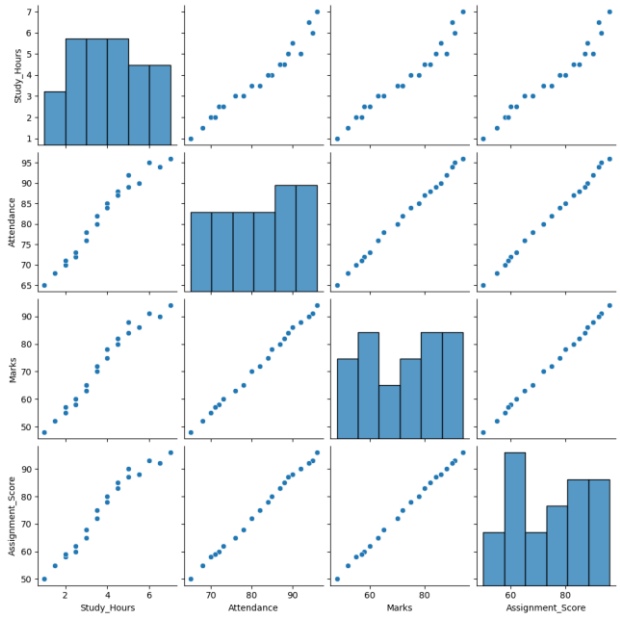


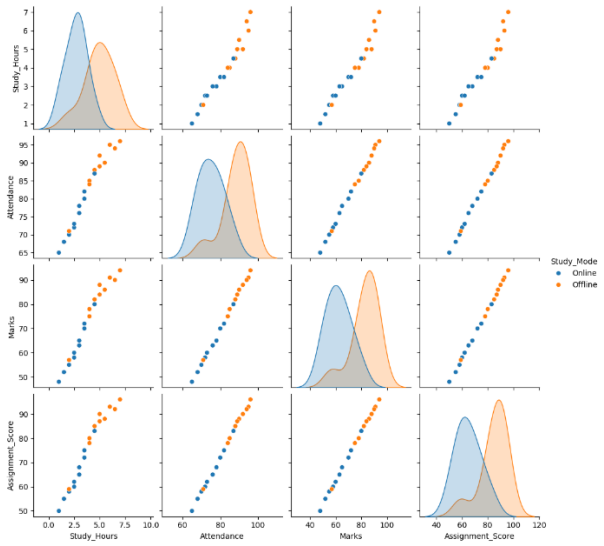


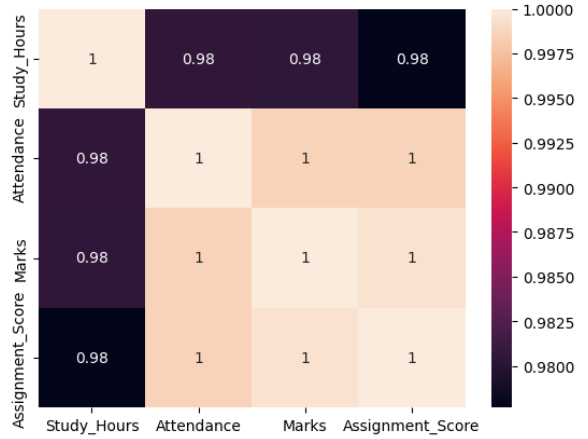

Marks
Study_Hours
100
90
80
70
60
50
1
2
3
4
5
6
7

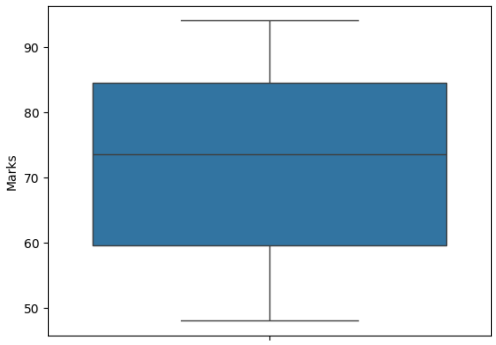

Marks
Study_Hours
90
80
70
60
50
1
2
3
4
5
6
7

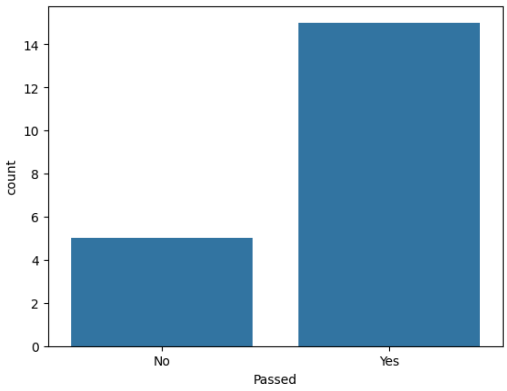

Male
50.0%
50.0%
Female

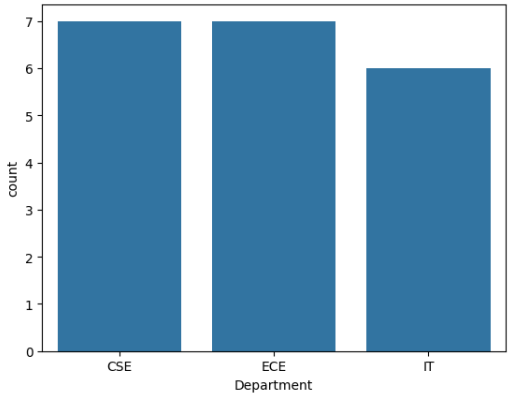

Study_Hours
Attendance
Marks
Assignment_Score

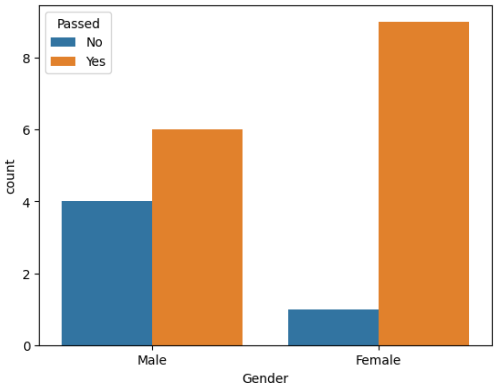

Study_Mode
Online
Offline
Study_Hours
Attendance
Marks
Assignment_Score

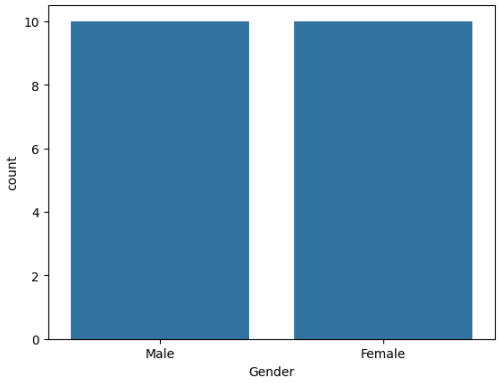

Study_Hours
Attendance
Marks
Assignment_Score
1
0.98
0.98
0.98
0.98
1
1
1
0.98
1
1
1
0.98
1
1
1
1.0000
0.9975
0.9950
0.9925
0.9900
0.9875
0.9850
0.9825
0.9800

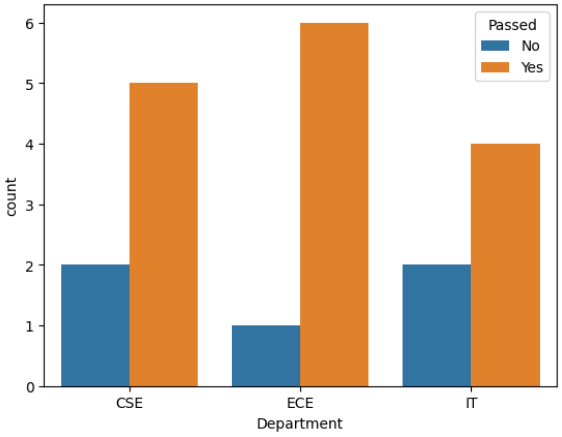












RESULT:

The given dataset is visualized by using python code.